



Indian Institute of Information Technology (IIIT) Surat
Department of Computer Science & Engineering
B.Tech-II, Semester - IV
Mid-Semester Examination, March-2025
Subject: Object Oriented Technology (CS 404)

Date: 6th March, 2025

Timing: 1:30 P.M to 3:00 P.M

Max Marks: 30

Instructions:

1. All parts of a question should be answered together.
2. Make suitable assumptions whenever necessary with proper justifications.
3. Figures to the right indicate maximum marks assigned for that question.

Q.1. Provide a detailed explanation of generalization, aggregation, and composition relationships. Then, classify the given relationships into one of the following categories: generalization, aggregation, composition, or association. [6]

- A software application consists of modules, and each module contains multiple functions.
- A music band consists of multiple musicians, and each musician plays one or more instruments.
- A bus route connects multiple cities, and each city has multiple bus stops.
- A city has multiple transportation systems, such as buses, metro, and trams. Each system operates under different rules and regulations.

Q.2. Prepare a UML Class Diagram for school from given description: "A school has a principal, many students, and many teachers. Each of these persons has a name, birthdate, and may borrow and return books. Teachers and the principal are both paid a salary; the principal evaluates the teachers. A school board supervises multiple schools and can hire and fire the principal for each school. A school has many playgrounds and rooms. A playground has many swings. Each room has many chairs and doors. Rooms include restrooms, classrooms, and the cafeteria. Each classroom has many computers and desks." [6]

Q.3. Design a UML Use Case Diagram for an *online frequent flyer system*, illustrating the functionalities available to both customers and airlines. Ensure that appropriate relationships between use cases are included where necessary. You can incorporate an abstract parent use case where applicable for generalization. The system provides the following functionalities: [6]

- **View Credits:** Display the current frequent flyer points in the account.
- **Submit Missing Credit:** Request credit for an activity that was not credited.
- **Change Address:** Update the mailing address.
- **Change User Name:** Modify the username associated with the account.
- **Change Password:** Update the account password.
- **Book Free Rewards:** Redeem frequent flyer points for: A free flight, A free hotel stay, and A free rental car
- ✓ **Request a Frequent Flyer Credit Card:** Apply for a credit card that offers flyer points as a reward.
- **Check Prices and Routes:** Search for possible flight routes and corresponding prices for paid travel.
- **Check Availability for a Free Flight:** Check availability of free travel for a specified flight.

Q.4. Design a UML State diagram for the smart door lock system. This system allows users to lock, unlock, and program the lock by entering a passcode. The system has three main operational modes: Locked, Unlocked, and Programming Mode. In Locked Mode, the door is secured. Entering the correct passcode switches it to Unlocked Mode. In Unlocked Mode, the door remains open until manually locked or automatically relocks after 30 seconds. In Programming Mode, users can change the passcode. Holding the "Set" button for 5 seconds enters this mode, where they can enter a new passcode. Illustrate all possible states and transitions, including incorrect password attempts, automatic relock, and programming mode timeout in the diagram.

Q.5. Answer any TWO of following questions. Each question carries 3 marks. [6]

(a) Discuss object oriented system development life cycle in detail with the help of block diagram.

(b) Discuss the main three models used in Rumbaugh object modeling technique in detail.

(c) Difference between structural approach and object oriented approach.

Best Wishes